DS5899: Special Topics in Data Science

Leveraging NLP in Asset Management

Week 3: Nvidia Hardware

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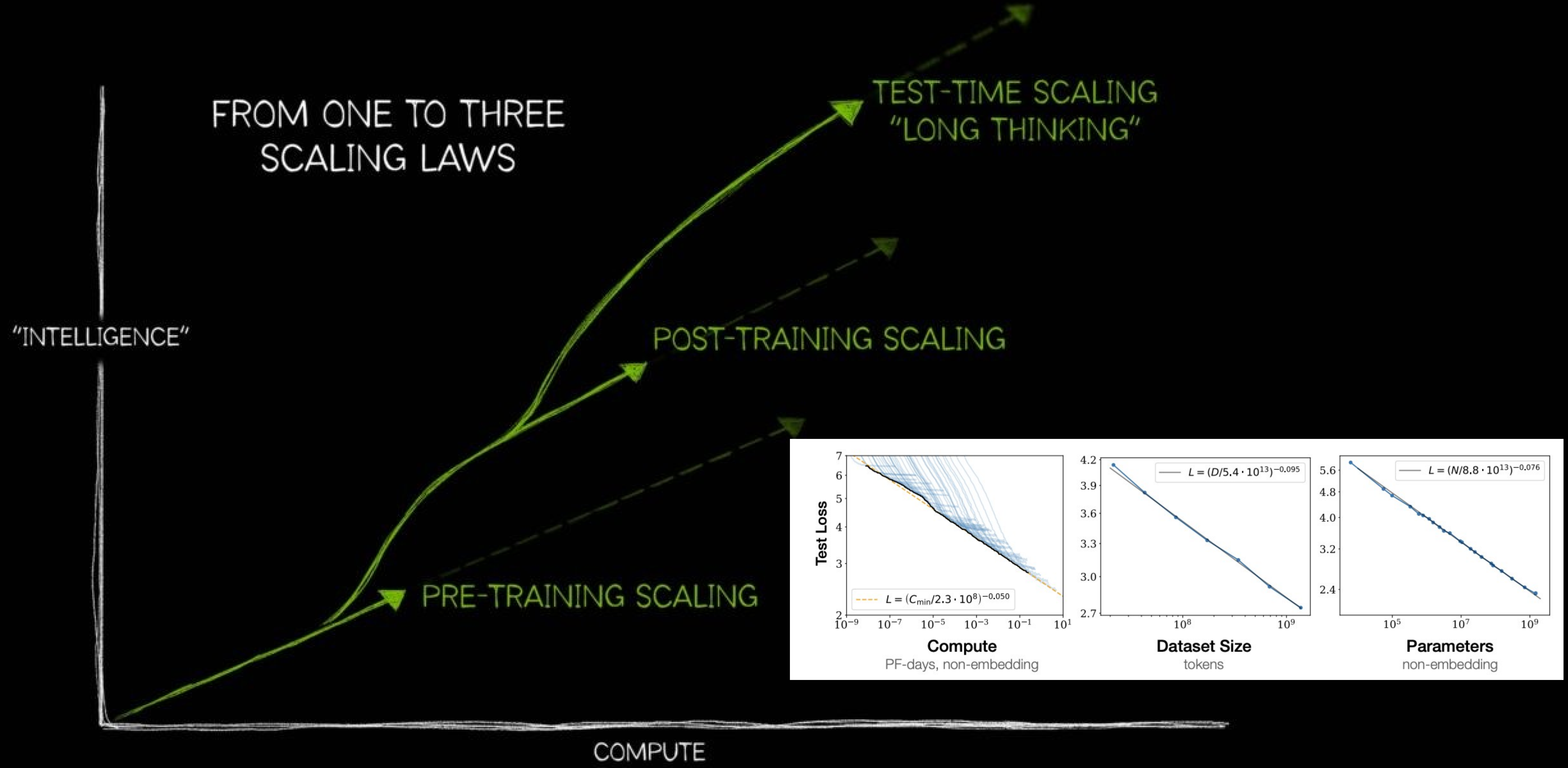
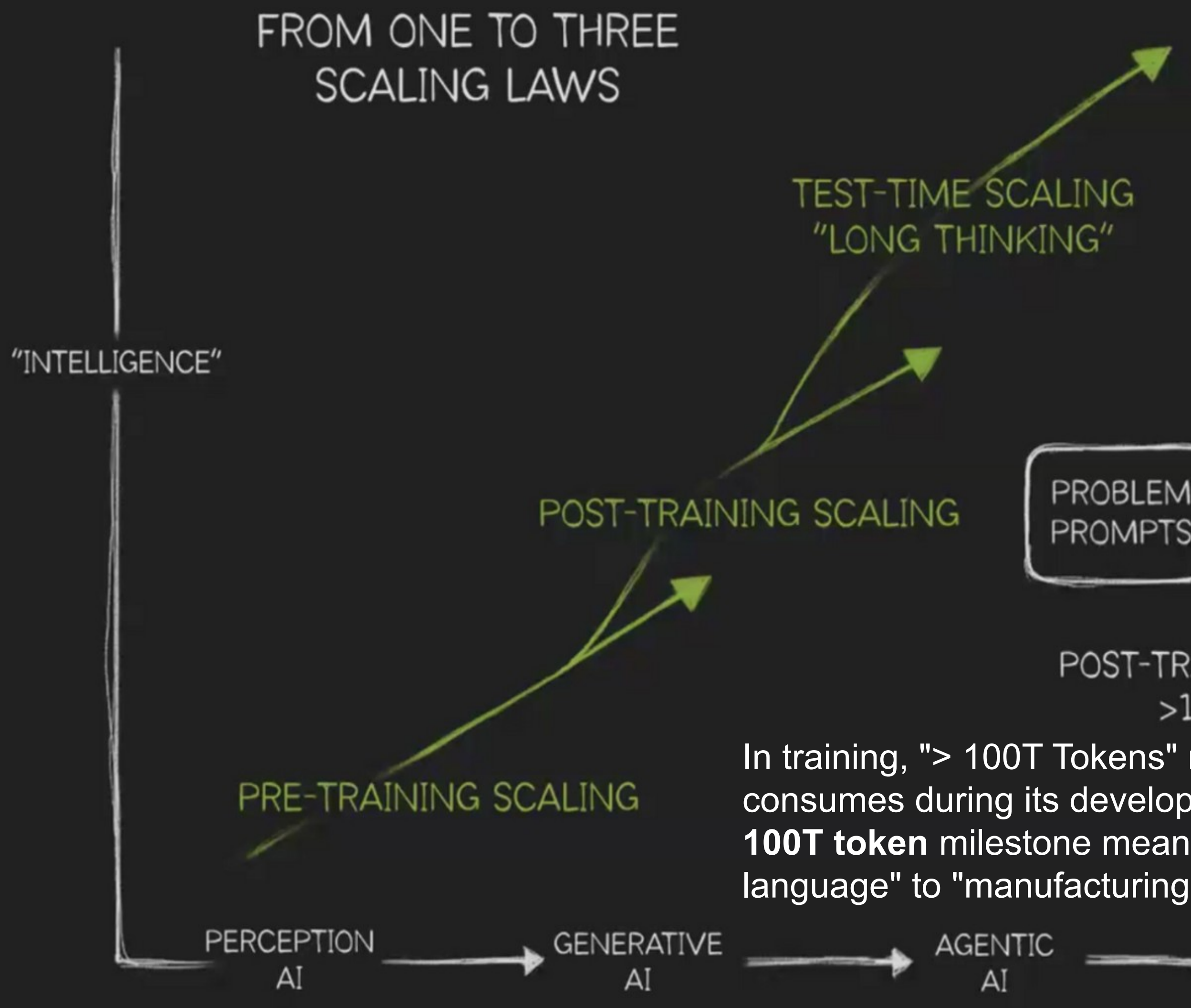
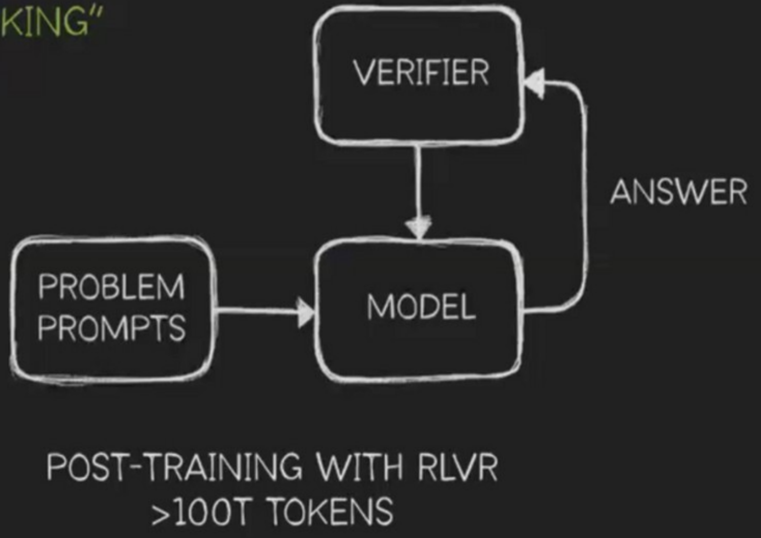
Perception AI Generative AI Agentic AI Physical AI

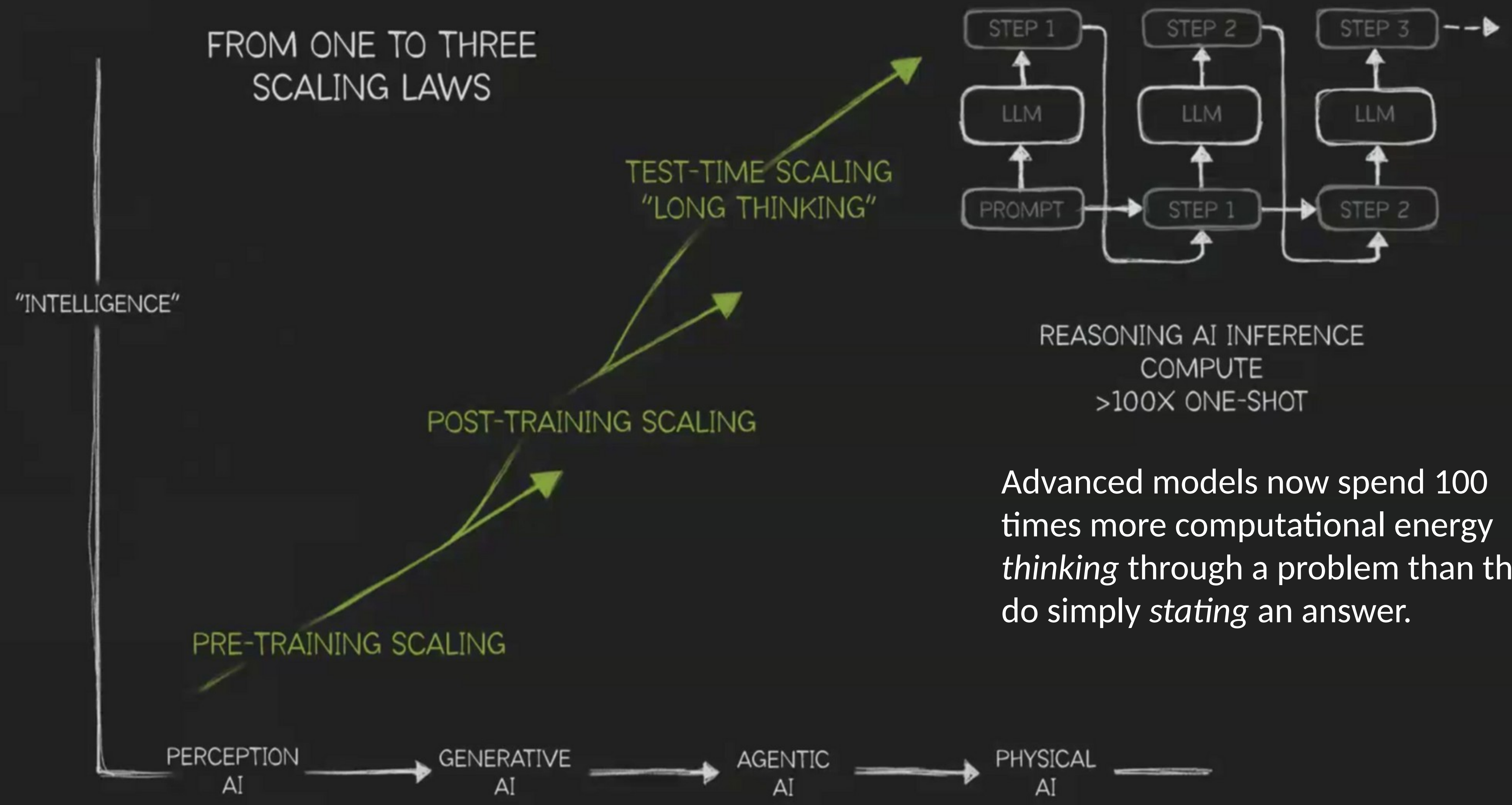




RLVR (**Reinforcement Learning with Verifiable Rewards**) is a training paradigm that replaces subjective human feedback with objective, programmatic "truth".



In training, "> 100T Tokens" refers to the total volume of data a frontier model consumes during its development. In your **AI Factory** framework, hitting the **100T token** milestone means the factory has transitioned from "learning the language" to "manufacturing intelligence" at an industrial scale.



Advanced models now spend 100 times more computational energy *thinking* through a problem than they do simply *stating* an answer.

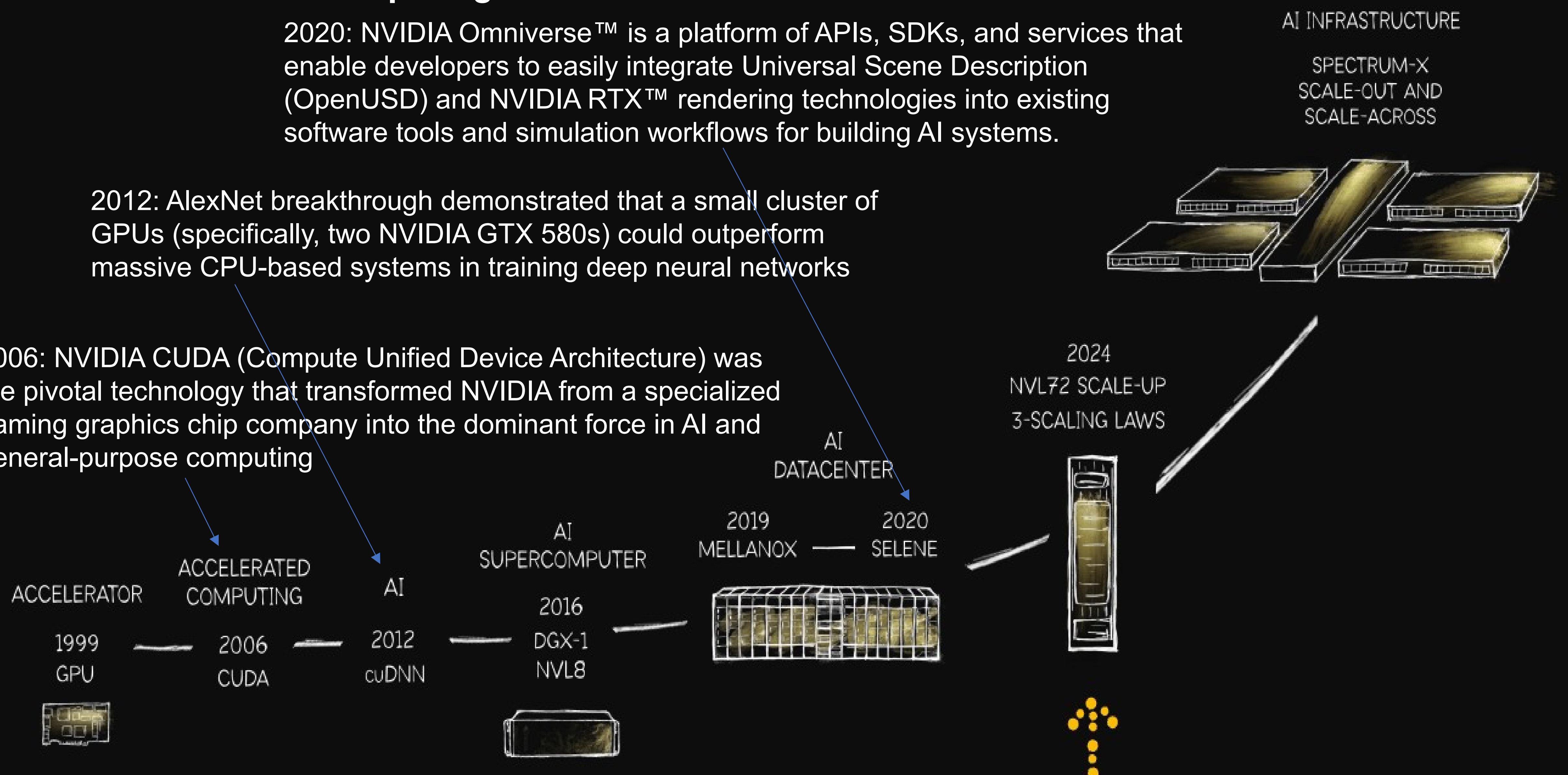
NVIDIA's Evolution From Chips to an AI Infrastructure Company

NVIDIA's (found in April 5, 1993) core business is **accelerated computing**

2020: NVIDIA Omniverse™ is a platform of APIs, SDKs, and services that enable developers to easily integrate Universal Scene Description (OpenUSD) and NVIDIA RTX™ rendering technologies into existing software tools and simulation workflows for building AI systems.

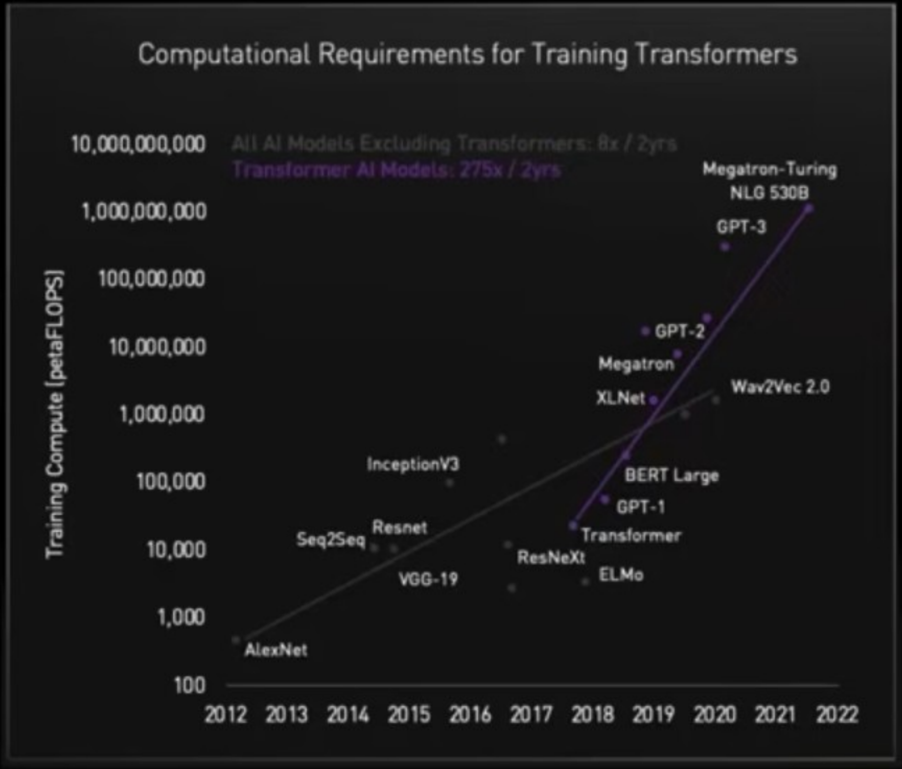
2012: AlexNet breakthrough demonstrated that a small cluster of GPUs (specifically, two NVIDIA GTX 580s) could outperform massive CPU-based systems in training deep neural networks

2006: NVIDIA CUDA (Compute Unified Device Architecture) was the pivotal technology that transformed NVIDIA from a specialized gaming graphics chip company into the dominant force in AI and general-purpose computing



Evolution of NVIDIA GPU Chips

(refer to Appendix for details)

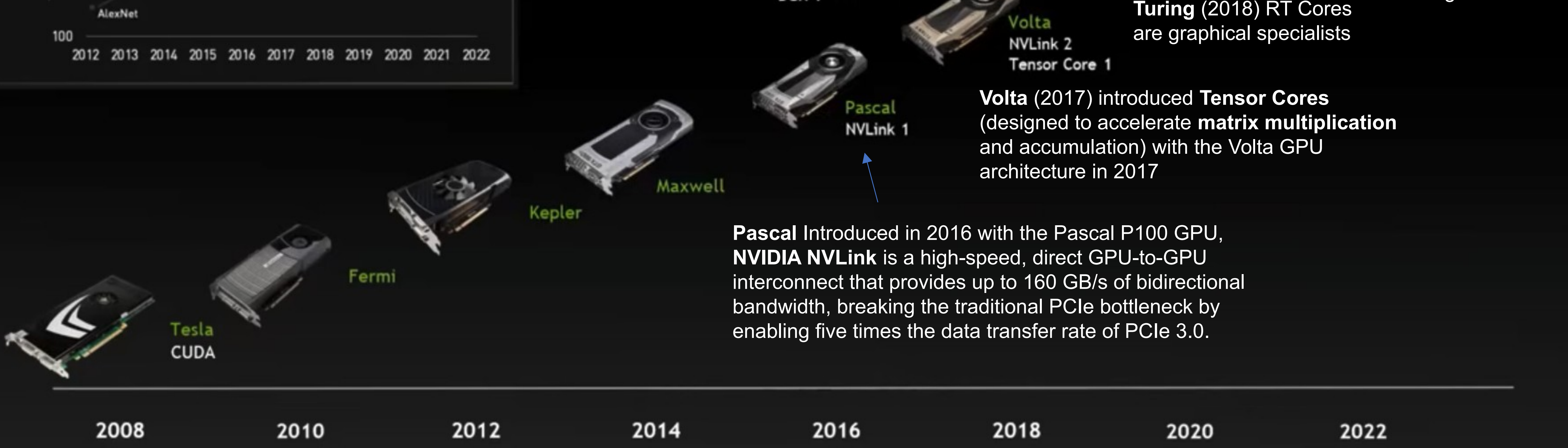


In 2016, Jensen Huang personally delivered NVIDIA's new "DGX Spark" AI supercomputer to Elon Musk at SpaceX's Starbase in Texas

DGX-1

Ampere (2020)

- 3rd Gen Tensor Cores: Introduced Tensor Float 32 (TF32)
- **MIG** allows a single Ampere GPU (like A100) to be partitioned into up to seven smaller, isolated GPU instances for different workloads



Ada Lovelace (2022) is for consumer gaming, professional visualization, and rendering

Turing (2018) RT Cores are graphical specialists

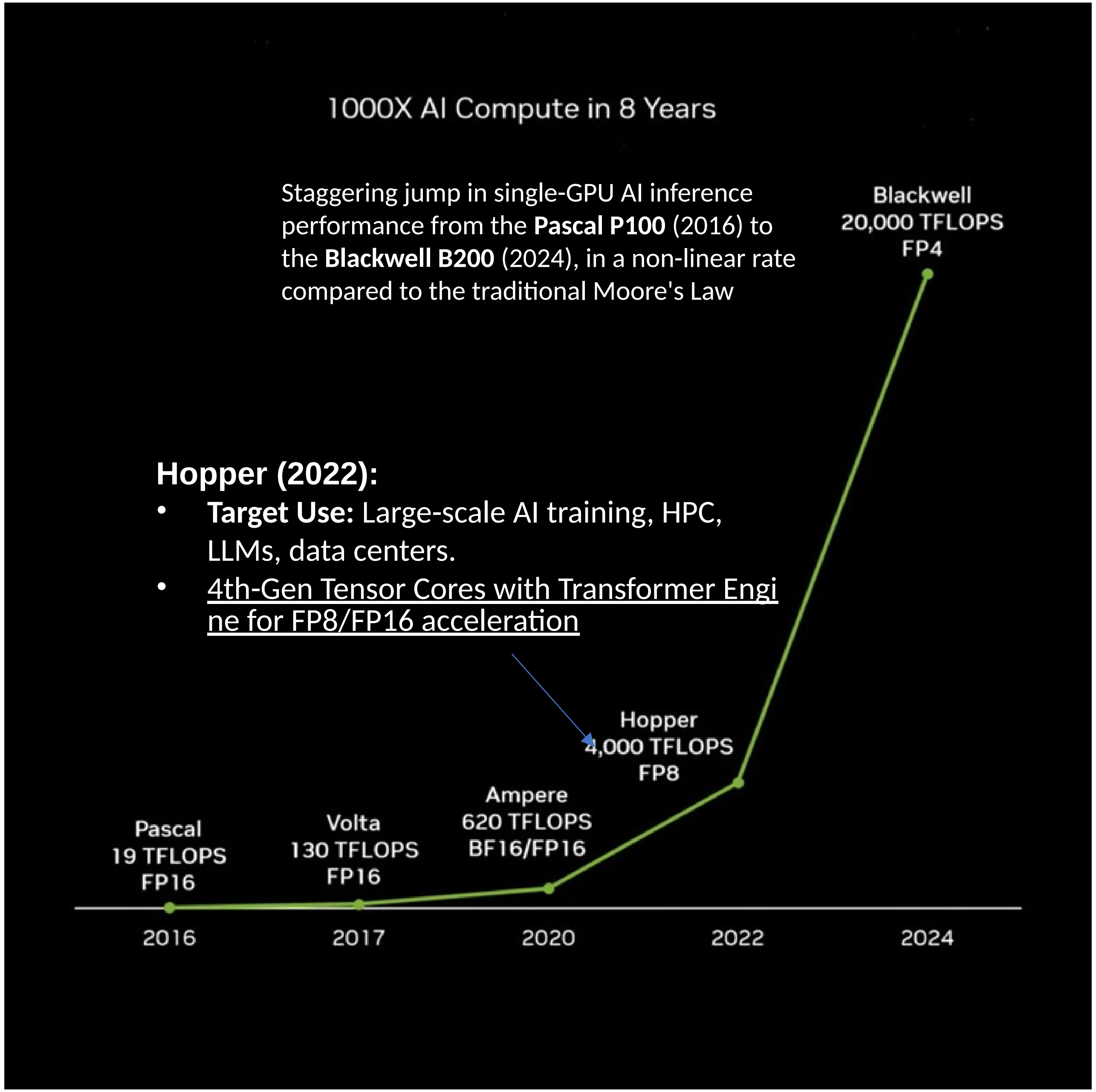
Volta (2017) introduced **Tensor Cores** (designed to accelerate **matrix multiplication** and accumulation) with the Volta GPU architecture in 2017

Pascal Introduced in 2016 with the Pascal P100 GPU, **NVIDIA NVLink** is a high-speed, direct GPU-to-GPU interconnect that provides up to 160 GB/s of bidirectional bandwidth, breaking the traditional PCIe bottleneck by enabling five times the data transfer rate of PCIe 3.0.

1000X AI Compute in 8 Years

- NLP Transformers became popular immediately following their introduction in **2017**, with widespread industry adoption and mainstream popularity occurring between **2018 and 2022**
- Hopper and Blackwell Key Features

Feature	Hopper (H100/H200)	Blackwell (B200/GB200)
Transistors	80 Billion	208 Billion
Best Precision	**FP8 (8-bit)**	**FP4 (4-bit)**
Design Type	Monolithic (Single Die)	Multi-Die (2 Dies)
New Hardware	Confidential Computing	Decompression Engine
Interconnect	**NVLink 4** (900 GB/s)	**NVLink 5** (1.8 TB/s)



NVIDIA Compute Cores

- **Cuda Cores** are designed to **run many tasks in parallel**. They're great for workloads like data preprocessing, simulations, video rendering, and traditional machine learning.
- **Tensor cores** are specialized for the **matrix operations** used in deep learning. They multiply and add blocks of numbers in a way that's far more efficient than CUDA cores can, especially when using mixed-precision formats like FP16, BF16, or INT8.
- **RT (Ray Tracing) Cores*** are dedicated hardware (ASICs) on RTX GPUs that accelerate the complex mathematical calculations for Ray Tracing, a rendering technique that simulates how light behaves to create photorealistic images with accurate shadows, reflections, and refractions in real-time.

*NVIDIA Omniverse relies on **RT Cores** within NVIDIA RTX GPUs to power its high-performance, photorealistic rendering capabilities, known as the **Omniverse RTX Renderer**.

The Evolution of Computing Paradigms

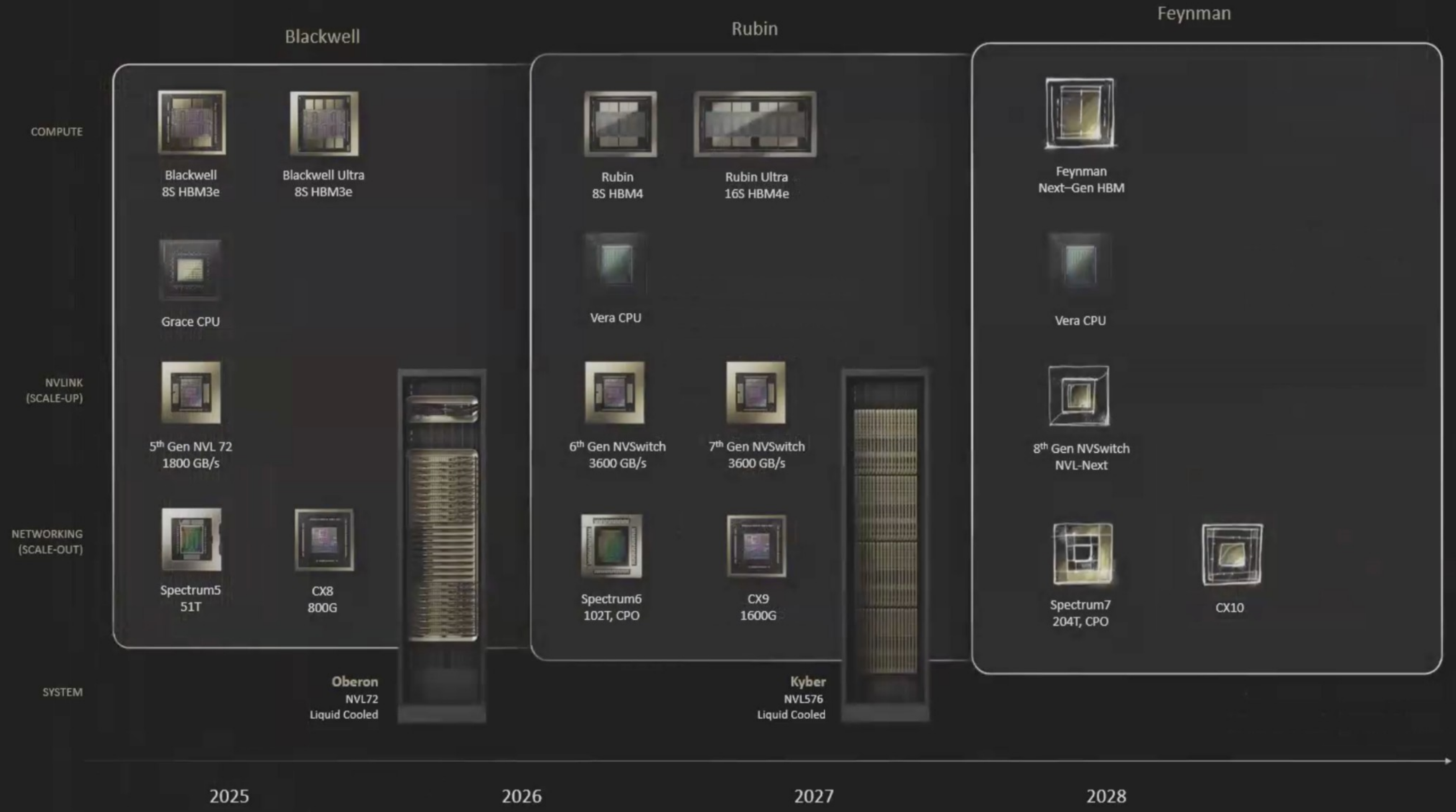
- Mainframe Era (1960s – 1990s): **Scale-Up**
  - Dominated by IBM and monolithic central processing.
- Distributed Era (2000s – 2020s): **Cloud Computing**
  - Driven by AWS and massive scale-out of commodity x86 servers.
- The AI Era (2020s – Present): **Hardware-Software Co-Design**
  - **HGX** e.g., Rubin NVL8 integrates 8 NVIDIA Rubin GPUs to function as one giant accelerator.
  - High-Speed Fabric: Leveraging **NVLink** for ultra-fast **GPU-to-GPU** memory sharing and **InfiniBand** for low-latency **Node-to-Node** networking to eliminate communication bottlenecks.
  - **DGX SuperPOD** (refer to Appendix for details) integrates HGX-based server systems along with high-speed networking, storage, and management software to create a turnkey, rack-scale AI infrastructure.

NVIDIA's Strategy: A Shift from Simple Scale-Out to High-Density Scale-Up

- The AI industry's focus is shifting from the "raw intelligence" of training to the operational productivity of inference. In this new paradigm, **AI factories** are essentially refineries:
  - You apply **ENERGY** as a raw material, and the factory produces **TOKENS** as a valuable commodity.
- While computational performance has increased a thousandfold, **energy supply** has emerged as the "master resource" and the most acute binding constraint on AI growth.
- To ensure that advanced intelligence remains economically viable within existing energy limits, for example, **NVIDIA Rubin** platform aims for a **10x reduction in inference token cost.**

NVIDIA Paves Road to Gigawatt AI Factories

One-Year Rhythm | Full-Stack | One Architecture | CUDA Everywhere



2 x Management top of rack Ethernet switches

4 x Power shelves

10 x Compute Nodes

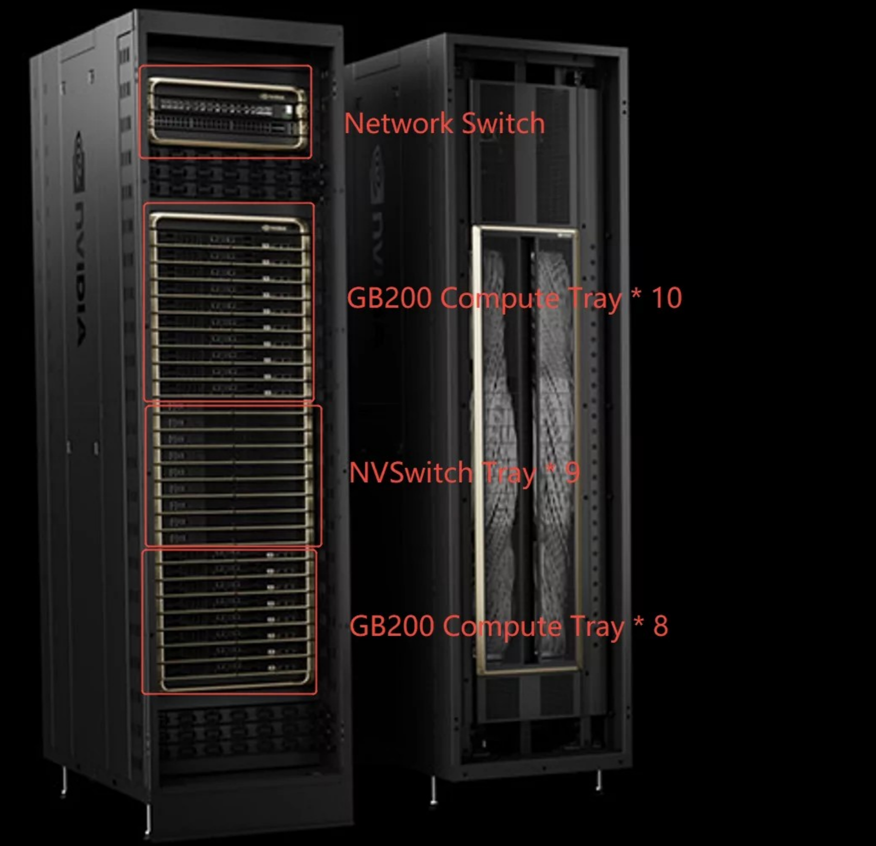
9 x NVLink switches

8 x Compute Nodes

4 x Power shelves

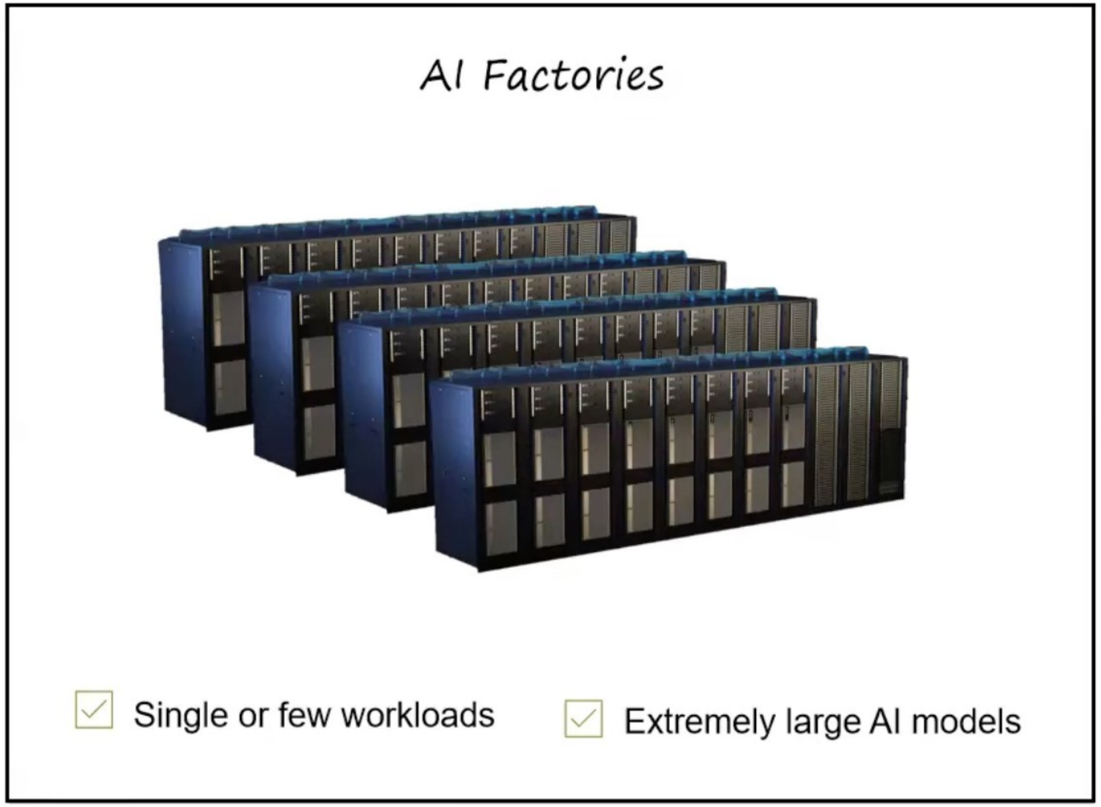
Seismic bracing

NVIDIA GB200 Server



The weight of an NVIDIA GB200 server system, specifically the rack-scaleNVL72 configuration, is substantial, reported around 1.36 tons or even 2 tons

Specialized Data Centers AI Factories versus AI Clouds



NVLINK and InfiniBand AI Fabric

Ethernet network

AI Factories handle massive, large-scale workflows and development of LLMs & other

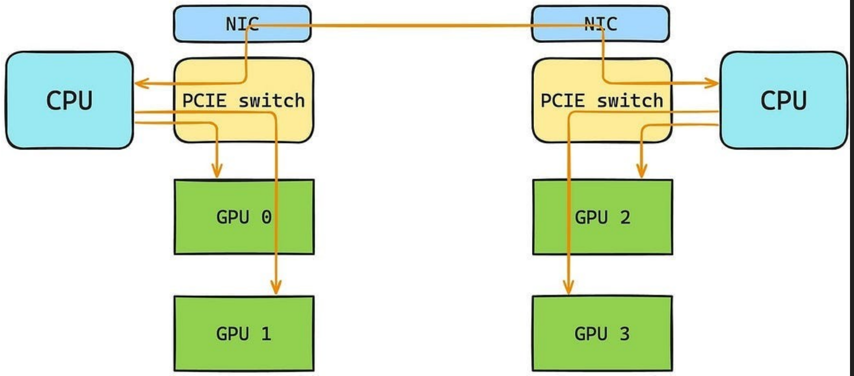
AI clouds extend the capabilities of traditional cloud infrastructure to support large-scale

NVIDIA Full-Scenario Compute Interconnects: From Intra-Rack to Inter-Data Center

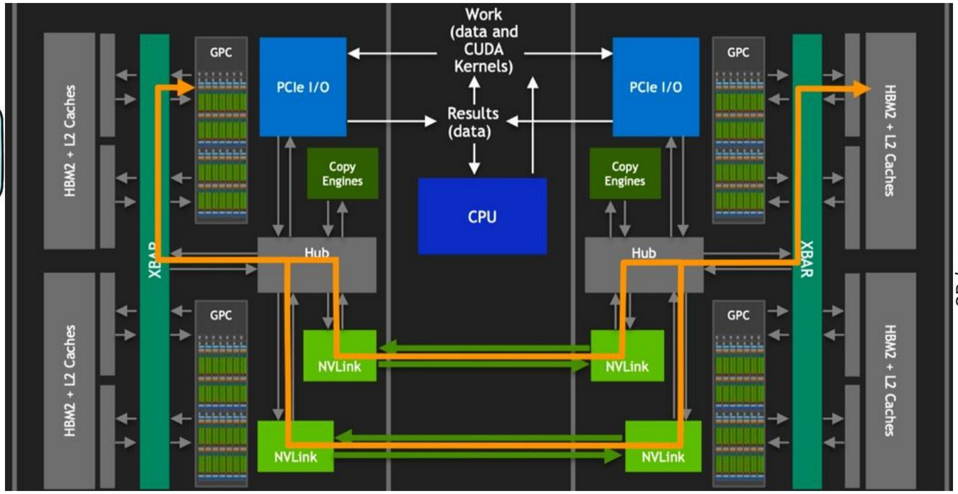
- **Intra-Rack Interconnect (Node-to-Node)**
  - Intra-rack interconnects primarily address communication requirements **within a single server rack.** It's a key component in enabling scale-up of the cluster.
- **Intra-DC(Data Center) Interconnect (Rack-to-Rack)**
  - Intra-DC interconnects are designed for the entire data center, typically covering hundreds to thousands of racks, these Intra-DC Interconnect links are used to build large-scale distributed computing clusters.
  - **Leaf-Spine** (switches) architecture has become the mainstream network form in modern DCs.
- **Inter-DC Interconnect (DC-to-DC)**
  - Inter-DC interconnects target geographically dispersed data centers, enabling DC to DC connectivity that spans ultra-long distances across cities and even regions.
  - Fiber optic links used for Inter-DC transmission over long distance inevitably introduce light-speed-level propagation delays.

NV Link Explained: The Key to Scaling Massive GPU Clusters

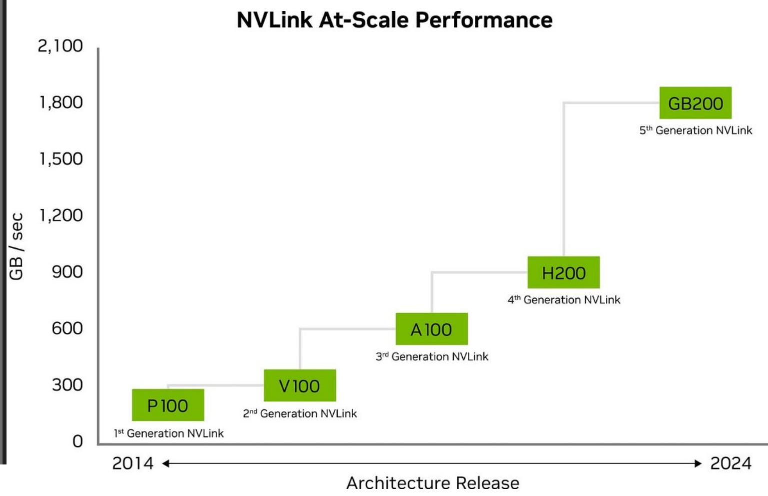
- Traditionally, building a high-performance compute node meant linking multiple GPUs directly to the CPU through a PCIe switch.
- Rather than replacing PCIe outright, NVLink is designed to complement it, handling the heavy lifting of GPU-to-GPU traffic while PCIe remains responsible for GPU-to-CPU and GPU-to-storage connections.



PCIe in Multi-GPU Systems



How NVLink Works



The evolution of NVLink

A Major Shift in AI Infrastructure, Moving from a GPU-centric to a Fabric-centric era

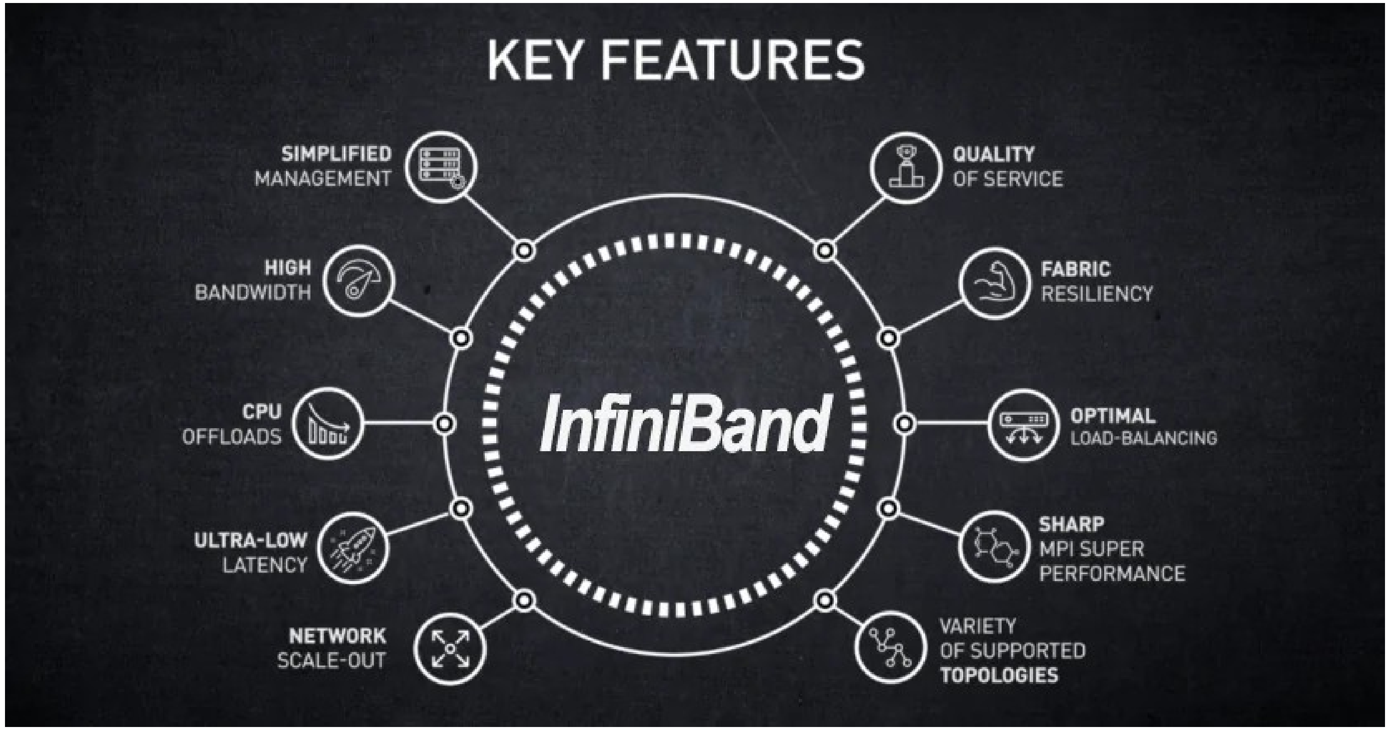
- **Generational Scaling**: 2 – 4 x FLOPS growth from Hopper to Blackwell to Rubin.
- **Balanced Architecture**: Proportional increases in HBM bandwidth & memory footprint.

Architecture	Peak AI Performance (FP4/FP8)	HBM Capacity	Bandwidth
Hopper (H200)	~4 PetaFLOPS (FP8)	141GB (HBM3e)	4.8 TB/s
Blackwell (B200)	~20 PetaFLOPS (FP4)	192GB (HBM3e)	8.0 TB/s
Rubin (R100)	~50 PetaFLOPS (FP4)	288GB (HBM4)	~13.0 TB/s

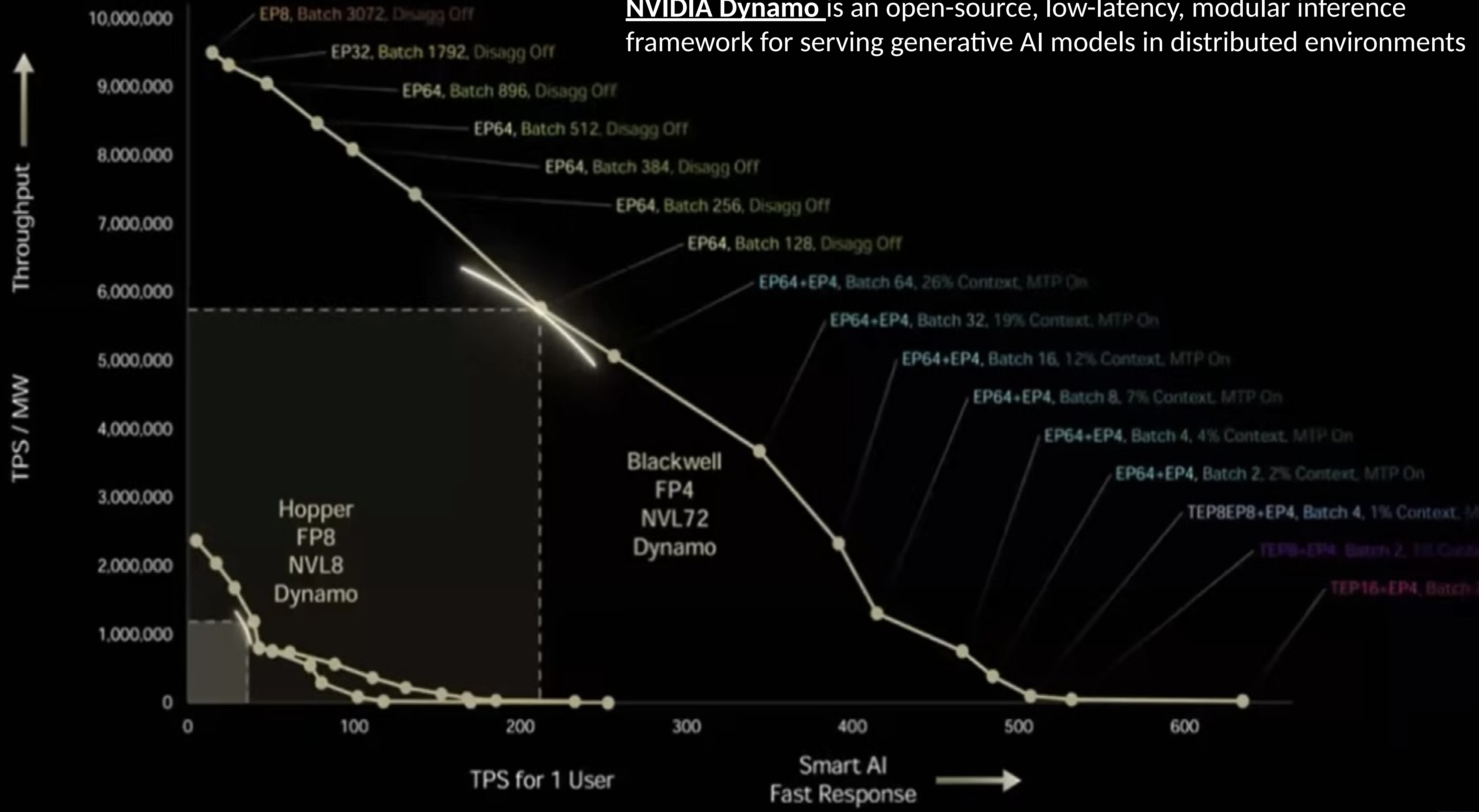
- One real **bottleneck** for training massive models (like GPT-4 or MoE architectures) is **interconnect** bandwidth, not the chips themselves.
- By opening NVLink, NVIDIA ensures that even if other companies (e.g., Hyperscaler) build their own chips (ASICs), those chips still reside within the NVIDIA-dominated network ecosystem (NVSwitch/NVLink), preventing GPUs from being excluded from the fabric.

InfiniBand (IB)

- **InfiniBand** is a high-speed interconnect technology designed for HPC and large-scale AI computing scenarios
- An IB Fabric is a Data Center network where servers and storage devices are interconnected through IB switches, forming a high-performance network.



<u>NVIDIA Dynamo</u> is an open-source, low-latency, modular inference framework for serving generative AI models in distributed environments



Key Trends in the AI & Token Economy from NVIDIA GTC Keynote 2026

- **Tokenomics Driving New Business Models**
  - The "Token Economy" is sparking a wave of business model innovation. We will see a surge of tools focused on the pricing, metering, and trading of tokens, mirroring the evolution of billing and optimization tools seen during the cloud computing era.
- **Agent Infrastructure**
  - Open-source projects like Open Claw are set to catalyze the development of enterprise-grade Agent tools. This includes specialized platforms for the development, deployment, security, and management of autonomous AI agents.

- **Data Acceleration**
  - Specialized libraries such as KUDF and KVS are shifting the landscape of the data processing market. Traditional data platforms must embrace GPU acceleration or risk becoming obsolete as high-performance data handling becomes the new standard.
- The Rise of Physical AI
  - For robotics startups, the "standard stack" will now include synthetic data simulation, world models, and robotics simulation platforms. These technologies are essential for bridging the gap between digital intelligence and physical execution.
- **Opportunities in Vertical Models**
  - NVIDIA’s open-model ecosystem is fueling the demand for industry-specific fine-tuning. This will give rise to a new category of "Sovereign AI" service providers who help organizations build and maintain their own customized, proprietary AI models.

Appendix: DGX SuperPOD

- DGX SuperPOD configurations typically utilize four distinct network fabrics to ensure high performance and efficiency for AI workloads
  - **Compute Fabric:** A high-performance, low-latency network (typically **InfiniBand**) designed for inter-node communication and scaling deep learning workloads, providing 800 Gbps (or higher) of full-duplex bandwidth.
  - **Storage Fabric:** A high-speed network designed for rapid, parallel I/O access to storage systems.
  - **In-Band Management Network:** An Ethernet-based network for traffic between compute nodes, management nodes, and external networks.
  - **Out-of-Band (OOB) Management Network:** An Ethernet-based network used for Baseboard Management Controller (BMC) and Intelligent Platform Management Interface (IPMI) monitoring.

Appendix: NVIDIA Data Center GPU Roadmap (2008 – 2026)

Architecture	Year	Key Products	Major Technological Milestone & Features
Tesla	2008	C1060, S1070	**Introduction of CUDA:** Enabled general-purpose GPU computing.
Fermi	2010	M2050, M2090	**ECC Memory Support:** Crucial for HPC reliability.
Kepler	2012	K20, K40, K80	**Dynamic Parallelism:** Improved GPU utilization.
Maxwell	2014	M4, M40, M60	**Energy Efficiency:** Massive jump in performance-per-watt.
Pascal	2016	P100, P4	**NVLink 1.0 & HBM2:** First high-speed GPU-to-GPU interconnect.
Volta	2017	V100	**1st Gen Tensor Cores:** Specialized units for AI matrix math.
Turing	2018	T4	**RT Cores:** Hardware-accelerated ray tracing; added INT8/INT4.
Ampere	2020	A100	**Multi-Instance GPU (MIG):** Hard hardware partitioning.
Hopper	2022	H100, H200	**Transformer Engine:** Dynamic FP8 precision for LLM training.
Blackwell	2024	B100, B200	**Dual-Die Design:** 208B transistors and FP4 precision support.
Rubin	2026	R100, Vera CPU	**HBM4 & NVLink 6:** 50 PFLOPS NVFP4 inference; 3.6 TB/s GPU bandwidth; integrates **Vera CPU**.